6 Polynomials

Definition 6.1.
A *polynomial* with coefficients in $\mathbb{Q}$ is an infinite sequence

$$(a_0, a_1, \ldots, a_n, \ldots) \qquad a_i \in \mathbb{Q}$$

satisfying the following: $\exists N \geq 0$ such that $a_i = 0, \forall i \geq N$.

Further

$$(a_0, a_1, \ldots, a_n, \ldots) = (b_0, b_1, \ldots, b_n, \ldots) \iff a_i = b_i \qquad \forall i \geq 0.$$

We define addition and multiplication of polynomials by the following rules:

$$(a_0, a_1, \ldots, a_n, \ldots) + (b_0, b_1, \ldots, b_n, \ldots) = (c_0, c_1, \ldots, c_n, \ldots)$$

where

$$c_i = a_i + b_i \qquad \forall i \geq 0$$

and

$$(a_0, a_1, \ldots, a_n, \ldots) \times (b_0, b_1, \ldots, b_n, \ldots) = (d_0, d_1, \ldots, d_n, \ldots)$$

where

$$d_i = \sum_{s+t=i} a_s b_t \qquad \forall i \geq 0.$$

Example 6.1.

$$(1, \frac{1}{2}, -\frac{1}{3}, 0, \ldots) + (0, 2, \frac{1}{4}, 1, 0, \ldots) = (1, \frac{5}{2}, -\frac{1}{12}, 1, 0, \ldots),$$

and

$$(1, 2, -1, 0, \ldots) \times (1, 3, 0, 0, \ldots) = (1, 5, 5, -3, \ldots).$$

Remark (Notations).

- Let $X = (0, 1, 0, \ldots)$ and $n \geq 1$ then

  $$X^n = X \times \ldots \times X = (0, 0, \ldots, 1, 0, \ldots)$$

  where the 1 is shifted to the n-th digit.
  Further for all $i, j \geq 0$

  $$X^i X^j = X^{i+j}$$

  (prove the above facts, for the first use induction on n).
- For $a \in \mathbb{Q}$ the polynomial

  $$a = (a, 0, \ldots, 0, \ldots)$$

  is called a *constant* polynomial and denoted simply a.
- Every polynomial $(a_0, a_1, \ldots, a_n, 0, \ldots)$, with $a_i = 0$ for all $i \geq n + 1$, can be written as

  $$(a_0, 0, \ldots) + (0, a_1, 0, \ldots) + (0, 0, a_2, 0, \ldots) + \ldots + (0, \ldots, 0, a_n, 0, \ldots)$$

  which; since

  $$(0, 0, \ldots, 0, a, 0 \ldots) = (a, 0, \ldots, 0, \ldots)(0, 0, \ldots, 0, 1, 0, \ldots),$$

equals

$$(a_0, 0, \ldots) + (a_1, 0, \ldots)(0, 1, 0, \ldots) + (a_2, 0, \ldots)(0, 0, 1, 0, \ldots) + \ldots + (a_n, 0, \ldots)(0, \ldots, 0, 1, 0, \ldots)$$
$$= a_0 + a_1(0, 1, 0, \ldots) + a_2(0, 0, 1, 0, \ldots) + \ldots + a_n(0, \ldots, 0, 1, 0, \ldots)$$

which we denote

$$f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n.$$

The $\{a_i\}_{0\leq i\leq n}$ are called the *coefficients* of f. If $a_n \neq 0$, then n is called the *degree* of the polynomial f and denoted $\deg(f)$. The coefficient a_n is called the *leading coefficient* of f, and a_0 is called the *constant coefficient* of f.

- We write

$$\mathbb{Q}[X] = \left\{ f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n \ : \ n \geq 0, \quad a_i \in \mathbb{Q} \right\}$$

and call it the set of polynomials with rational coefficients. Identifying the set of constant polynomials with $\mathbb{Q}$ we can view $\mathbb{Q}$ as a subset of $\mathbb{Q}[X]$.

- The constant polynomial $a_0 = 0$ is called the *zero* polynomial.

Note that $\mathbb{Q}[X]$ is endowed with two operations $+$ and $\times$ as defined above which are the addition and multiplication of polynomials.

Lemma 6.1.
Given polynomials $f, g \in \mathbb{Q}[X]$ we have

$$\deg(f + g) \leq \max(\deg(f), \deg(g)),$$

with equality if $\deg(f) \neq \deg(g)$, and

$$\deg(f \times g) = \deg(f) + \deg(g)$$

if f and g are both non zero polynomials.

Proof. Exercise. □

Proposition 6.2.
$(\mathbb{Q}[X], +)$ is an abelian group.

Proof. First it follows from the definition of addition of polynomials that it is commutative since the addition of rationals is. Associativity of addition of polynomials also follows easily from the associativity of addition of rationals. The neutral element is the constant polynomial $0 = (0, 0, \ldots, 0, \ldots)$, also called the zero polynomial: $f(X) + 0 = f(X)$, $\forall f \in \mathbb{Q}[X]$. Further, given $f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n$, then $-f(X) = (-a_0) + (-a_1)X + (-a_2)X^2 + \ldots + (-a_n)X^n$ satisfies $f(X) + (-f(X)) = 0$ hence is the inverse of $f(X)$. □

Definition 6.2.
Let $f, g \in \mathbb{Q}[X]$. We say that g divides f if $\exists h \in \mathbb{Q}[X]$ such that

$$f(X) = g(X)h(X).$$

Example 6.2.
The polynomial $1+X+X^2$ divides $2+3X+3X^2+X^3$ since $(2+X)(1+X+X^2) = 2+3X+3X^2+X^3$.

Lemma 6.3.
Let $u \in \mathbb{Q}[X]$. Then u divides the constant polynomial 1 if and only if u is a non zero constant polynomial, i.e. $u \in \mathbb{Q} \setminus \{0\}$. Such u is called a unit of $\mathbb{Q}[X]$.

Proof. Assume $0 \neq u = a_0 \in \mathbb{Q}$ is a non zero constant polynomial. Let $v = \frac{1}{a_0}$ then $uv = 1$ and u divides 1.

Conversely let $f(X) = a_0 + a_1X + \ldots a_nX^n$ be a unit with $n = \deg(f)$, i.e. $a_n \neq 0$. There exists a polynomial $g(X) = b_0 + b_1X + \ldots b_mX^m$ of degree m; thus $b_m \neq 0$, such that $f(X)g(X) = a_0b_0 + \ldots + a_nb_mX^{n+m} = 1$.

In particular $a_nb_m \neq 0$ and $\deg(f(X)g(X)) = n + m = \deg(1) = 0$. Hence $n = m = 0$ and $u = a_0 \neq 0$ since $a_0b_0 = 1 \neq 0$. □

Definition 6.3.
Let $f \in \mathbb{Q}[X]$ be a non constant polynomial, i.e. $f \in \mathbb{Q}[X] \setminus \mathbb{Q}$ hence $\deg(f) \geq 1$. The notation $\mathbb{Q}[X] \setminus \mathbb{Q}$ means those elements in $\mathbb{Q}[X]$ which are not in $\mathbb{Q}$.

We say that f is *irreducible* if whenever $f(X) = g(X)h(X)$, where $g(X)$ and $h(X)$ are polynomials in $\mathbb{Q}[X]$, then either g or h is a unit, i.e. either g or h is a non zero constant polynomial.

In other words, f is irreducible if $f(X)$ doesn't factor as the product $g(X)h(X)$ of two polynomials in $\mathbb{Q}[X]$ with $\deg(g(X)) \geq 1$ and $\deg(h(X)) \geq 1$.

Example 6.3.

1. A polynomial $f = a_0 + a_1X \in \mathbb{Q}[X]$ of degree 1 (thus $a_1 \neq 0$) is necessarily irreducible. Indeed $f = gh$ implies $\deg(g) = 0$ or $\deg(h) = 0$, as $\deg(f) = 1 = \deg(g) + \deg(h)$. Thus either g, or h, is a constant polynomial, and is non zero (otherwise $f = gh = 0$). Hence either g or h is a unit.
2. The polynomial $X^2 - 3$ is irreducible. Indeed assume

$$X^2 - 3 = (aX + b)(cX + d) = acX^2 + (ad + bc)X + bd$$

with a and c both non zero.
Equating we get $ac = 1$, $ad + bc = 0$, and $bd = -3$. Hence $\{ad + bc = \frac{1}{c}d - \frac{3}{d}c = 0\} \Longrightarrow \{d^2 - 3c^2 = 0\} \Longrightarrow \{d^2 = 3c^2\} \Longrightarrow \{d = c = 0\}$, where the last implication follows from the fact that 3 is not a square of a rational number. But this contradicts $ac = 1$.

Theorem 6.4 (Euclidean division. The division algorithm).
Let $f, g \in \mathbb{Q}[X]$ with $g \neq 0$. Then $\exists h, r \in \mathbb{Q}[X]$ unique such that

$$f = hg + r$$

where either $r = 0$ (in which case g divides f), or $r \neq 0$ and $\deg(r) < \deg(g)$.

Proof. Write $f(X) = a_0 + a_1X + \ldots + a_mX^m$, $g(X) = b_0 + b_1X + \ldots + b_nX^n$ with $a_m \neq 0$ and $b_n \neq 0$. Assume for simplicity that $m \geq n$ and set $d = m - n \geq 0$ [if $m < n$ take $h(X) = 0$ and $r(X) = f(X)$)]. We argue by induction on d.

If $d = 0$, i.e. $m = n$, then $f(X) = \frac{a_m}{b_n} g(X) + r(X)$ where $r(X) = f(X) - \frac{a_m}{b_n} g(X) = a_0 + \dots a_m X^m - b_0 \frac{a_m}{b_n} - \dots - \frac{a_m b_n}{b_n} X^n$ which has degree $< n$.

Assume now that $k = d > 0$ and the theorem holds for $0 \leq d \leq k-1$. Set $f_1(X) = f(X) - \frac{a_m}{b_n} X^{m-n} g$. then $\deg(f_1) < \deg(f)$ (check this fact). If $f_1 = 0$ then $f(X) = \frac{a_m}{b_n} X^{m-n} g(X)$ and the theorem holds with $h(X) = \frac{a_m}{b_n} X^{m-n}$ and $r(X) = 0$. If $f_1(X) \neq 0$ then by the induction hypothesis $f_1(X) = h_1(X)g(X) + r_1(X)$ where either $r_1(X) = 0$ or $r_1 \neq 0$ and $\deg(r_1) < \deg(g)$. Thus $f(X) = h_1(X)g(X) + r_1(X) + \frac{a_m}{b_n} X^{m-n} g(X) = (h_1(X) + \frac{a_m}{b_n} X^{m-n}) g(X) + r_1(X)$ and the theorem holds with $h(X) = h_1(X) + \frac{a_m}{b_n} X^{m-n}$ and $r(X) = r_1(X)$.

Next, we show the unicity of h and r. Assume $f(X) = g(X)h(X) + r(X) = g(X)h'(X) + r'(X)$ with (h, r) and (h', r') satisfying the conclusion of the Theorem. If $r = 0$ then $f = gh = g'h' + r'$ hence $g(h - h') = r'$ which implies $r' = 0$ [otherwise $\deg(r') \geq \deg(g)$. Likewise $r' = 0$ implies $r = 0$]. Assume both r and r' are non zero, in which case $r = f - gh$ and $r' = f - gh'$ have both degree less than $\deg(g)$. The difference of these two polynomials $r - r' = gh' - gh = g(h' - h)$ has also degree less than that of g. If $h' - h = 0$ then $r' - r = 0$ and we have $h = h'$, $r = r'$. On the other hand $\deg(r - r') = \deg(g) + \deg(h - h')$ if $h - h'$ is a non zero polynomial, which contradicts $\deg(r - r') < \deg(g)$. Thus $h - h' = 0$ and $r - r' = 0$ necessarily which implies $h = h'$ and $r = r'$. □

Example 6.4.
Let $f(X) = X^6 + 2X^5 + X^4 + 3X$ and $g(X) = X^2 + 1$. Then $X^6 + 2X^5 + X^4 + 3X = (X^4 + 2X^3 - 2X)(X^2 + 1) + 5X$, thus $h(X) = X^4 + 2X^3 - 2X$ and $r(X) = 5X$ in this case.

Theorem 6.5 (Unique Factorisation of Polynomials).
Let $f(X) \in \mathbb{Q}[X]$ be a non zero polynomial which is not a unit, i.e. $f(X) \in \mathbb{Q}[X] \setminus \mathbb{Q}$, or equivalently $\deg(f(X)) \geq 1$.

Then it holds

$$f = g_1 g_2 \dots g_r$$

(product of polynomials) with g_i irreducible for all $1 \leq i \leq r$. Further if

$$f = h_1 h_2 \dots h_s$$

is another such factorisation, with h_i irreducible for $1 \leq i \leq s$, then $r = s$ and after rearranging the $\{g_i\}$ one has $g_i = u_i h_i$ where u_i is a unit.

Proof. The existence of a factorisation $f = g_1 g_2 \dots g_r$ as above follows by induction on the degree of f: if f is irreducible then we are done. Otherwise $f = qm$ where $q, m \in \mathbb{Q}[X]$ are non zero, non units, with $1 \leq \deg(q) < \deg(f)$ and $1 \leq \deg(m) < \deg(f)$. By induction both q and m have a factorisation as a product of irreducible polynomials which induces a similar factorisation for f.

The second assertion follows from the fact that if an irreducible polynomial h divides the product $f_1 f_2$ of two polynomials then either h divides f_1 or h divides f_2 (more details on the uniqueness of factorisation will be seen in §10). □

Definition 6.4.

1. Let $f(X) = a_0 + a_1 X + \dots + a_n X^n \in \mathbb{Q}[X]$ and $c \in \mathbb{Q}$. We define the *value* $f(c)$ of f at c by

$$f(c) = a_0 + a_1 c + \dots + a_n c^n \in \mathbb{Q}.$$

Thus f can be viewed as a function

$$f : \mathbb{Q} \to \mathbb{Q}$$

defined by

$$c \mapsto f(c).$$

If $g \in \mathbb{Q}[X]$ is another polynomial then (check this fact)

$$(f \times g)(c) = f(c)g(c), \quad (f + g)(c) = f(c) + g(c).$$

2. We say $c \in \mathbb{Q}$ is a *root* of f if $f(c) = 0$. In this case we say f has a root in $\mathbb{Q}$.

Example 6.5.

1. A polynomial $f(X) = a_0 + a_1X \in \mathbb{Q}[X]$ of degree 1 (thus $a_1 \neq 0$) has necessarily a root in $\mathbb{Q}$. Indeed $f(-\frac{a_0}{a_1}) = 0$ and $-\frac{a_0}{a_1}$ is a root of f.
2. The polynomial $X^2 - 2$ has no root in $\mathbb{Q}$: a root would be equal either to $\sqrt{2}$ or $-\sqrt{2}$ which are both not rational numbers.

Lemma 6.6.
Let $f \in \mathbb{Q}[X]$, $\deg(f) \geq 1$. If $c \in \mathbb{Q}$ is a root of f then the polynomial $X - c$ divides f.

Proof. By the division algorithm $\exists h, r \in \mathbb{Q}[X]$, unique, with

$$f(X) = h(X)(X - c) + r(X)$$

and either $r = 0$ or $\deg(r) < \deg(X - c) = 1$. Thus $r \in \mathbb{Q}$ is a constant.

Further

$$f(c) = h(c)(c - c) + r = 0 + r = 0$$

hence $r = 0$, and $f(X) = h(X)(X - c)$. □

Lemma 6.7.
Let $f \in \mathbb{Q}[X]$ with $\deg(f) = n \geq 1$. Then f has at most n roots in $\mathbb{Q}$.

Proof. We argue by induction on $n = \deg(f)$. If $n = 1$ then $f(X) = a_0 + a_1X$ and $-\frac{a_0}{a_1}$ is the only root of f.

Suppose the lemma is true for polynomials of degree $< n$. Assume $c \in \mathbb{Q}$ is a root of f. Then by Lemma 6.6 there exists $g \in \mathbb{Q}[X]$ with

$$f(X) = (X - c)g(X)$$

and $\deg g = n - 1$. Assume $c' \neq c$, $c' \in \mathbb{Q}$, is another root of f. Then

$$f(c') = (c' - c)g(c') = 0,$$

hence $g(c') = 0$ necessarily and c' is a root of g since $c - c' \neq 0$. By induction there are at most $n - 1$ such c', hence at most n roots of f. □

In a similar way we could have defined the set of polynomials $\mathbb{C}[X]$ with coefficients in $\mathbb{C}$ (and indeed in any field, see §7 for the definition of a field)

$$\mathbb{C}[X] = \{f(X) = a_0 + a_1X + a_2X^2 + \ldots + a_nX^n \ : \ n \geq 0, \ a_i \in \mathbb{C}\}.$$

Polynomials with coefficients in $\mathbb{C}$ share several of the properties of polynomials in $\mathbb{Q}[X]$: addition, multiplication, Euclidean division, unique factorisation. The following result is fundamental in Algebra.

Theorem 6.8 (Fundamental Theorem of Algebra)**.**
Let $f \in \mathbb{C}[X]$ *be a polynomial with* $\deg(f) \geq 1$. *Then* $\exists c \in \mathbb{C}$ *with* $f(c) = 0$.